

Malware Analysis Lab – Answer Sheet

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Task 1 – Static Analysis (PEStudio 9.51)

Instructions: Open **hello.exe** in PEStudio and record the following:

Architecture: PE32 + 64 bit executable

Entry Point (EP): 0x00021AA0

Number of Imports: 3 imports (libraries)

Key Imports (list 2–3)

- MessageBoxA
- GetProcAddress
- VirtualProtect

Sections (VA, Virtual Size, Raw Size / Offset, Permissions)

- UPX0 — VA: 0x00001000, Virtual Size: 0x00019000, Raw Size/Offset: 0x00000000 / 0x00000200, Permissions: Writable, Executable, Self-modifying, Virtualized
- UPX1 — VA: 0x0001A000, Virtual Size: 0x00008000, Raw Size/Offset: 0x00007E00 / 0x00000200, Permissions: Executable, Self-modifying
- UPX2 — VA: 0x00022000, Virtual Size: 0x00001000, Raw Size/Offset: 0x00000200 / 0x00008000, Permissions: Executable

Task 2 – Dynamic/Hybrid Analysis (VirusTotal)

Instructions: Upload the file or SHA256 hash to VirusTotal and record observations.

Detection Ratio: 42/72

Example AV Names

- Microsoft - Trojan:Win32/Wacatac.B!ml
- Alibaba - Trojan:Win64/CobaltStrike.1dde65b8
- Yandex - Trojan.GenAsa!OVbAfmTNtYI

MITRE Tactics and Techniques

- **Execution**
- **Defense Evasion**
- **Command and Control**

What tool was used to pack this software for defensive evasion?

Packer: UPX (4.24) [NRV,brute]

Task 3 – Reflection

Instructions: Summarize your findings in 3–5 sentences, comparing static and dynamic analysis.

Reflection:

From the static analysis in PEStudio, I could tell the file was a 64-bit executable that was packed using UPX, especially because of the section names like UPX0, UPX1, and UPX2. It also showed signs of self-modifying behavior and had very few imports, which usually means it's trying to hide what it's doing. When I looked at it in VirusTotal, a large number of vendors flagged it as malicious and identified it as a Trojan. The behavior details showed things like unpacking itself at runtime and modifying memory, which are common malware techniques. Overall, static analysis helped me spot the red flags in the file structure, while dynamic analysis confirmed that it was actually malicious.